

Chapter 13

How Museums Made (and Re-Made) Their Digital User

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Abstract Surveying archival material from across fifty years of practice, and leveraging multi-disciplinary theoretical perspectives from digital studies, design studies and disability studies, this chapter uses a form of ideational history, to evidence shifting linguistic habits and frames of reference around the construct of the ‘digital user’, at once both sequential and enduring: from the user as ‘operator’ (set within a system-orientated context of organizational efficiency); to the user as ‘individual’ (amidst priorities of usability and experience design); to – emerging today – the user as ‘actant’ (against which social value and agency are the new indices of success). In these different constructions, the ‘digital user’ moves from being part of the system, to outside system, to in the world, whilst its principal capability shifts from automation, to personalization to empowerment. Crucially, it is proposed here that it has been the role of accessible design (in particular the design of digital with disability in mind) that has acted both as a key informant and agitator of this change, but also the practice through which this development can most usefully be seen.

Introduction

Museums have been talking about computers for over fifty years. And as they have talked, so their specialist vocabularies have become ‘registers of experience’ (Agnew, 2014, p. 312), collecting their own heritage. ‘Interactive’, ‘audio-guide’, ‘virtual tour’, ‘online collection’, ‘digital engagement’: these terms (and many like them) all carry forward assumptions regarding technology, the museum and their audiences. Consequently, to look at this lexicon used by the museum profession over time, is to trace a history of digital practice. Indeed, terms such as ‘user’ and ‘digital user’ may at first seem perfunctory. And yet, by looking back – deeply and carefully – at how these words and concepts have been constructed and deployed by the museum (and commentators on the museum), we can bring into focus changing priorities, values and intellectual informants. We can, in short, through an ideation history such as this, start to see a sector revealed through its re-making of a word.

Surveying half a century of practice from archival material of three institutional collections (a national museum institution, a sector support agency and an advocating cultural charity), and leveraging multi-disciplinary theoretical perspectives from digital studies, design studies and disability studies, this chapter analyses what Klein (2011) would call the ‘linguistic habits’ of the museum computing community – specifically around the use of the terms ‘user’ and ‘digital user’. In general terms, the aim here is to animate the value of an ideational approach to the history of digital cultural heritage and to demonstrate the potential of a more expansive criticality around the relationship between ‘museum’ and ‘digital’. More specifically, the objective here is to evidence an evolving frame of reference around the construct of the ‘digital user’. It is through this critical long view, that we can witness the user positioned initially as ‘operator’ (part of the system); to then a user constructed as ‘individual’ (outside of the system); to – emerging today – the user acknowledged as ‘actant’ (active ‘in the world’). Concurrently, in these different constructions, we notice the principal capability of the ‘digital user’ shifting from ‘automation’ (set within a 1970s–80s system-orientated context of organizational efficiency); to ‘personalization’ (amidst 1990s–2000s priorities of usability and experience design); to finally ‘empowerment’ (against which social value and agency are the new indices of success).

What emerges is a usage of the term, at the ‘semantic core’ (Froeyman, 2014) of digital practice in the sector, shaped in part by technological change, but also from a growing professional ethic of reflective practice and accessible provision. Crucially, it is proposed here that accessible design (particularly the design of digital with disability in mind) has acted both as a key informant and agitator of this change, but also the practice through which this development can most usefully be seen. The discussion here proposes not just how the concept of the museum’s ‘digital user’ has remained in motion, but, moreover, that this motion (this making, and re-making of the term) serves witness to changing and evolving conceptualizations around accessible digital design, and, indeed, around the very idea of the museum itself. We are left, ultimately with an original modelling of ‘user’ (and ‘use’) that is not only a means through which to understand the sector’s past but is also – *in the era of the postdigital, medialization and digital purposefulness* – a means to articulate our on-going practice.

Tracing a ‘History of Use’

Fittingly, for a study that intends to leverage a historical long view, it is towards the archive (rather than to, say, modern design and use) that this investigation looks. Specifically, the research here draws upon the collections of the Smithsonian Institution Archives (in the USA), as well as the organizational archives of the Collections Trust (in the UK) and the UK-based charity for accessible digital culture – the Jodi Mattes Trust. Within the Smithsonian Institution Archives it is, specifically, the extensive holdings relating to the organization’s ‘automation’ and computerization that provide some of the most significant insights into changing language and terminology around digital use.

In contrast, the Collections Trust archive holds over 3,000 documents spanning four decades – including the organization’s earlier incarnations as ‘mda’, then before that ‘The Museum Documentation Association’, and back to its origins as the ‘Information Retrieval Group of the Museums Association’. With the remit of the Trust (and its earlier formations) centering on supporting standards (and skills around those standards) in collections and information management (Collections Trust, 2018), its archive of manuals, reports and guides, helps expose our story of shifting assumptions around ‘use’ and ‘user’. From the Jodi Mattes Trust, it is the archive relating specifically to the charity’s running (since 2005) of the ‘Jodi Awards’ that is of particular import here. The archive of nominations and the careful, evidenced, structured cases that each make for distinctive and innovative practice warranting recognition, can all be seen as a ‘prism’ (Parry, 2018) through which to see an evolving relationship within the museum sector, between design, digital and disability. In the language of the winners, their citations and their reflections on practice, we hear assumptions around the digital user, consciously set against the Trust’s ambition for a world in which ‘disabled people hold both equal access to information and cultural experience through and within digital media, and equal opportunity as participants, creators, co-creators, artists and employees in the cultural sector’ (Jodi Mattes Trust, 2018).

As a triptych, these three collections complete not only a useful chronology (the Smithsonian Institution Archives reaching back to the 1960s; the Collections Trust strongly focused on the 1980s and 1990s; the Jodi Mattes Trust Archive exclusively relating to 2000s and 2010s onwards), but they also build for us a horizon of both published and unpublished material – evidencing both public-facing and internal discourse around the ‘digital user’. It is from their various minutes, applications, project briefs, reports and manuals, that we can hear the authentic voices of both practitioners and policy makers alike, the day-to-day language of work alongside the rhetoric of outward communication.

As we handle these collections, and this evidence, we do so mindful of a wider critical context. Specifically, we recognize here how other disciplines continue to attempt their own anatomies of the ‘user’ and the ‘digital user’. Outside of museology, we note how this terminology has, for instance, undergone considerable scrutiny in media and communication studies. This includes highly self-reflective discourse on what Schröder (2013, p. 49), at least, sees as the complementary methods and paradigms for understanding ‘audiences and users’; as well as scholarship that has tracked the turn towards a more ‘user-centric rather than a technology-centric perspective’ on the practice of design – the shift to thinking more expansively about ‘contexts of user and user goals’ (Kunert, 2009, p. vii). These methodological and the practical perspectives on ‘use’ and ‘user’, have also extended to more philosophical questioning of the relationship between media and the human, in a world in which, according to Lagerkvist (2017, p. 97) ‘[q]uestions concerning digital technologies are [...] challenges about human existence’. This is scholarship that emphasizes ‘situated use’ and the larger ‘socio-cultural context’ of digital design – such as Participatory Design and its commitment to enhancing users’ ‘autonomy’ and ‘power of decision’, enabling users to ‘understand the systems they are part of’ (Stuedahl, Morrison, Mörtberg & Bratteteig, 2010, p. 3, 6). Within this community of research that continues to follow the characteristics and consequences of changing digital users and uses of digital design, it is John Hartley’s visionary eye that, most notably, chooses to register ‘the emergence [...] of the ‘user’, as opposed to the consumer’, in the list of timely motives for rethinking the ‘underlying model of communication’. Confidently and circumspectly, Hartley (2012, pp. 2–3) surveys the transformation in the intellectual framework of media and cultural studies ‘from a linear to a dialogic mode; from producer to consumer; from powerful corporation and charismatic celebrity to everybody in the population; from representation to productivity; from structural opposition to dynamic systems; from cultural studies to cultural science’. Hartley’s declaration (and challenge) to his subject is to turn from what he calls a ‘provider’ perspective to a ‘user’ one; for him ‘the audience’ has transmogrified into ‘the user’ (Hartley, 2012, pp. 8, 114).

Looking up from museum studies, and away from media and communication studies, we are struck by the multiple parallel conversations that continue in other parts of the academy around the ‘user’ and the ‘digital user’. We see computer science plotting the ‘evolution’ of user-centered design through to UX, and the common centering in each of the ‘user’, ‘usability’ and ‘prototype use’ (Knight, 2011, p. 10) – what Noyes and Baber (1999, p. x) powerfully brand as ‘the stance [...] to design systems with the focus primarily on the user’. In the

context of data visualization, we see GIS articulating a shift in emphasis to a more ‘user centred perspective for map design’ (Jones, 2010, p. 42). And we see critical management studies calling out their own ‘turning point’ in the use of IT, specifically the alignment of business success with the requirement of ‘putting the human perspective – the digital user – at the very heart of the new digitized service-led economy’ (Brenner et al., 2014); this is a new orthodoxy of (re-)designing corporate processes ‘from the ground up, starting with the user’s perspective, thus putting usage experience and utility of the individual center stage’ (Brenner et al., 2014).

As ever for the museum, it is perhaps developments within library and information studies (as a cognate discipline and an adjacent profession) that provide particularly resonant inspiration and orientation. We notice, for instance, that as a discipline library science has already framed its own ‘history of the “user-centered paradigm”’, and been prepared to classify the shift from the ‘system’ to the ‘user’ point of view as a ‘revolution’ (Bowler et al., 2011). As the library sector attempts to understand its role when, according to some of its commentators and practitioners, ‘the traditional information environment is breaking down’, so the focus turns to understanding who their users are and what their new expectations might now be (Baker & Hall, 2013, pp. 2–3). Notably for our discussion here (and its thesis around the contemporary museum user as socially engaged ‘actant’) we hear a language in this library studies research that foregrounds a ‘predominance’, ‘power’ and ‘control’ of the information user (Baker & Hall, 2013, pp. 5–6). There is a user with ‘prerogatives’ (Frosio, 2012, p. 5) and ‘privileges’ (Guibault, 2012, pp. 91, 105).

From the methodological, practical and philosophical recasting of the ‘user’ in media and communication studies, through to the centering of the user’s point of view in computer science and business studies, and the rise of the empowered user in information studies, we see multiple disciplines sustaining their own conversations around the definition, the experience and the role of the ‘user’ and the ‘digital user’. However, whilst sensitive to this context (and, indeed, partly inspired and guided by it) our discussion here, nonetheless, attempts to contribute something different. The outcomes of this study may contribute to a shared sensibility to the significances of the term ‘user’ in a way that will inform wider practice and scholarship alike. It is also intended to provide an example – dare one say, a ‘use case’ – to communication and media studies (and other disciplines) on how these core concepts and discussions have played out in the unique mediatized space of the museum. In other words, our intention here is both to draw from, but then in turn to feed respectively back into (and inform) the media scholarship around the ‘user’, using our discrete museological perspective. This means back into these other academic conversations and sub-fields of ‘user, use & utility research’ (Brenner et al., 2014); into those live discussions around ‘discretionary use’ (Knight, 2011, p. 10) and the freedom and choice users have; as well as challenging the HCI and UCD orthodoxy that the user is subject to other ‘influences’, be that ‘workplace’ or ‘environment’ (Noyes & Babar, 1999), and instead suggest that the user, rather, has agency in the world. Similarly, the ‘actant’ identified here (another persona for the user, alongside their earlier, but persisting, identities as ‘operator’ and ‘individual’) belongs with more recent calls within media studies for what Lagerkvist (2017, p. 107) enthralling calls a ‘change of casting’ and a greater sensibility to the sociality and the fundamental humanness of the ‘principal inhabitant’ of the media terrain.

To trace our ‘history of use’, we also need one final other element of analytical equipment to put alongside the reliable substantive *archive* and this wider critical *context* – and this relates to the *assumptions* being made here around scholarly and professional language. After all, the argument set out here is predicated upon the idea that language (terms such as ‘user’ and ‘digital user’) passes on cultural information. Thinking fundamentally about language, theoretical evolutionary biology shows us how language comes from ‘shared attention, shared intentionality’, and evolves in a ‘highly social, potentially cooperative context’ (Szathmáry & Számadó, 2008). Moreover, historical linguistics provides the framework to allow for historicity within our reflection of professional and specialist terms (Vennemann, 2002, p. 245), as well as a precedent for tracing the evolution of linguistic features. Ethnolinguistically, Görlach (2002, p. 83) gives us ways of assembling histories of terms in other sectors and industries – such as advertising. Likewise, we see writers such as Kerwin Lee Klein (2011), tracing the development of terms such as ‘theory’, ‘memory’ and ‘historiography’ within the discipline of history – what later readers have seen as ‘a book about words [...] those words that have formed the semantic core of our thinking about history’ (Froeyman, 2014).

Similarly, in the area of human geography, writers such as John Agnew (2014), have ‘decoded’ some of his subject’s ‘important geographical concepts [...] according to the norms of the history of ideas, broadly construed’. In his case, Agnew helps us specifically to see a tension within what he terms an ‘idealist epistemology’, namely the idea that the formal meaning ascribed to a word in a discipline or area of practice cannot be presumed to be the limit on how that word is used in all circumstances by that subject. His warning – *for those within his subject and for those of us outside* – is of a ‘naïve realist empiricism’ that uses these specialist and professional terms without thought to their genealogy (Agnew, 2014, p. 311). As if speaking directly to our project here, Agnew advises how ‘[t]racing the meanings of such words historically is valuable insofar as it can alert us to how some, often highly selective, meanings have changed’ (Agnew, 2014). Most importantly for Agnew, however, are the

practices and behaviors that sit around the use of these words. As useful as a ‘history of ideas’ might be, his concern is the risk of becoming ‘obsessed with tracing the intellectual genealogies’, detached from the ‘practical lives’ of those that might be related to them (Agnew, 2014, p. 318). With this caution held close to us, we therefore approach our own ‘history of ideas’ here on the ‘digital user’, mindful of the need to resist abstraction, and to ground this reading and this analysis firmly within professional practice and evidence.

We see a similar tracing of language and terminology taking place in the field of design and design history, not least as the term ‘design’ itself has expanded to ‘include almost any type of transformative activity’ (Huppatz, 2011, p. 262). Indeed, almost overwhelmed by what some commentators have seen as the ‘vastness and ubiquity of design culture’ (Willis, 2008, p. 162), and a ‘a sprawling and saturating phenomenon’ (Highmore, 2009), the task of compiling a dictionary of design risks becoming ‘an impossible task, one of an infinite regression of terms [...] liable to dissolve into babbling incoherence’ (Willis, 2008, p. 162). And so, it is here, confronted with such a challenge around academic terms, that the intellectual influence of Raymond Williams is perhaps most important to recognize – in particular, his reflections on specialist language, and his analysis of specialist language within academia.

First published in 1976, Williams’ canonical ‘Keywords: A Vocabulary of Culture and Society’, is self-fashioned as ‘the record of an inquiry into a vocabulary: a shared body of words and meanings concerned’ in this specific case with ‘the practices and institutions described as ‘culture’ and ‘society’’ (Williams, 2013, p. 13). The work was revisited in 2005 by sociologist and cultural theorist Tony Bennett, along with Lawrence Grossberg, and Meaghan Morris, introducing multiple authors and a new overtly international perspective (Bennett et al., 2005). Williams inspires our approach here to trace the history of the museum’s ‘digital user’, in a number of ways. First, Williams attempted to set out the meanings associated with the common words used within his subject areas, but not owned and contained solely within these disciplines; a helpful, and encouraging, context in which to reflect on ‘user’. Second, he sees these meanings as ‘developing’ and that the connections around them for the community of use was both ‘explicit’ and ‘implicit’. And, thirdly, he approaches these words (and shares the histories of their changing meaning and use) as both a way of showing how these concepts were both discussed and seen:

I began to see this experience as a problem of vocabulary, in two senses: the available and developing meanings of known words, which needed to be set down; and the explicit but as often implicit connections which people were making, in what seemed to me, again and again, particular formations of meaning – ways not only of discussing but of seeing many of our central experiences. (Williams, 2013, p. 13).

In short, Williams helps us to find a ‘a history and complexity of meanings’ as he calls it, that sits beyond the dictionary definition (Williams, 2013, p. 15). Finally, the sense that he writes at a point ‘where several disciplines converge but in general do not meet’ (Williams, 2013) is recognizable not just to those of us that work in the precinct of digital cultural heritage and museum computing, but to anyone within the wider field of museum studies – a subject area characterized by its concurrent, catholic, multiplicity of disciplinary perspectives.

To summarize, our analytical approach here is to take a diachronic (rather than synchronic) approach to the study of the terms ‘user’ and ‘digital user’ – choosing to plot changing usages and consequences of these terms. Underpinning this approach is an ‘ideational theory of meaning’ – that the meanings of words are subjective ideas. And, consequently, we consider the terms here as potential ‘registers of experience’ (Agnew 2014, p. 312; Wickberg, 2001, p. 384). But as we do, and as we recognize that these words and terms are also a piece of ‘shared intentionality’ and ‘shared attention’, we also assume a stance of ‘dynamic nominalism’ – acknowledging that these changing contexts may also only partly explain the things they describe (Szathmáry & Számadó, 2008). It is this socio-linguistic perspective (not usual, it should be stressed, for digital cultural heritage as a subject) that, as Froeyman (2014) so beautifully puts it, ‘increases our sensibility to the implications of seemingly innocent words’.

The User as ‘Operator’

The first set of linguistic habits we notice from the professional archive is around the ‘user’ constructed as an ‘operator’ of a machine, within a defined and closed system. This is the ‘user’ type (and use of the term) that particularly characterizes the early years of computerization within the sector – a time when computers were first being noticed as an ‘extremely useful tool’ for museums (McCormick, 1965, p. 281). Carefully briefing the Board of Regents of the Smithsonian Institution with a primer on this emerging technology, an annual report from 1965 records with measured excitement how ‘computers are acquiring much faster operating speeds and that their storage capacity is increasing while at the same time their physical size is decreasing [...] Many more thousands

of people in the next few years will find that digital computers will play an essential part in their activities' (McCormick, 1965, p. 298). For these early adopters and these early constructions of the computer 'user', this 'use' related specifically to designing new systems, new forms of programming and identifying requirements for data processing for making collections information more accessible, in ways hitherto unseen in the industry (Squires, 1969, pp. 3–6). In these contexts, we see the 'user' (not yet termed a 'digital user') typically located within what are seen as institution-wide (if not sector-wide) technological challenges.

Moreover, the 'user' referred to is often (if not always) generic; the new wave of recommendations for novel information resources that marked these first two decades of museum computing are distinctive in the way they identify 'users' in general terms. Whilst there may be references to broad categories of 'scientists/curators', 'specialists' or 'other scholars' (Lytle, 1981, pp. 2, 11), the generalization is – more often – sustained at the level of 'a staff function' (Lytle, 1981, p. 11) or simply 'human factors', largely seen in terms of other systems (Smithsonian Institution, 1982, p. 2). As well as generic, this 'operator' is also largely anonymous.

'The computer does not know', one early professional report reads, 'whether the sequence of instructions which it performs were written by an accountant, linguist, philosopher, librarian, theologian, social scientist, physical scientist, engineer, or mathematician' (McCormick, 1965, p. 299). The operations of the system are, as this report puts it, 'unified' into a single set of uses by a single set of users. Most strikingly, this is a 'user' that is defined entirely in terms of the system of which they are considered part – be that the 'collections information system', 'catalog information system', 'inventory system', 'scheduling system', or the 'mailing system' (Smithsonian Institution, 1982, pp. 4–7). The utility of the system is measured in how the 'user' and machine operate together; one recommendation from an early 'automation' report captures this characteristic, suggesting that planned 'user experience trials' ought to identify 'present and projected types of request for information' (Squires, 1969, p. 32). Here the 'user of the system' (Squires, 1969) is a component of that system, the test being of the operation of operator and machine together.

Notably, this generic, anonymous 'user' (part of the system) is presented as operating within a culture of efficiency and operational effectiveness. This is the 'cost effective use of technology to support defined program needs' (Smithsonian Institution, 1982, p. 1), using technology as a 'tool that would permit doing a large number of operations to solve a problem' (McCormick, 1965, 298), of the museum's computer user using technology to identifying collections information 'more precisely' or by 'discerning more accurately their relationship to human society' (Lytle, 1981, p. 1). The effectiveness of these new technologies (and users of technology) are, in this context of 'use', gauged around their 'utility' (Squires, 1969, p. 30) for specific functions within defined systems: project management software 'is used to control the status of the [museum's] internal projects'; '[i]nventory control is used in various ways to keep updated lists of various collections'; 'a financial control database is used by the administrative services office to keep tabs on museum federal and trust funds'; 'word processing software [...] used to support certain collections registration functions' (Smithsonian Institution, 1984, p. 17). Recurrently, we see this early discourse of 'use' centered on the machine and its functionality.

The User as 'Individual'

By the early 1980s, museums were over a decade into their move to computerization, and – with it – came other (alternative) linguistic formulations around 'use'. As they began to review and reiterate their policies around data processing and information resource management, and as they started to develop more audience-facing interfaces for the technology, we hear in the professional discourse new nuances around the construction of the 'user' – particularly with respect to *context*. One key policy and planning report from this time, for example, refers to how: 'Curators would like to be able to sit at their terminals to plan an exhibit, key in several concepts which define the scope and purpose of the planned exhibit, and retrieve a listing and pictures of objects' (Bearman, 1982, p. 3). Compared to the discourse of the 'operator' from a decade (or more) before, here, crucially, our 'digital user' has gained *situation*. Here the museum is attempting to construct the 'user' empathetically, considering the curator's perspective and the curator's place (both influentially and physically) in the institution. Significantly, in this alternative construction of the 'user' (distinct from the systems operator) there is now an acknowledgement of a space beyond of the system – and, importantly, that the user can occupy a place within that space. For instance, in its choice of language we see the profession actively attempting to acknowledge the situated 'needs' of varied users, and aiming 'to solicit comments from interested parties', be that inside or outside of the organization (Smithsonian Institution, 1984, p. 1). 'The environment must permit controlled access to outside resources', one internal discussion suggests, as well as providing access to the museum's information 'by outside users' (Smithsonian Institution, 1982, Appendix B).

We also notice in this alternative construction of ‘digital users’ that they are no longer generic and anonymous (like the ‘operator’), but instead they are identified as ‘individuals’ – who might have their own varied propositions on use and design that the museum ‘has not considered or acquisition options it has not anticipated’ (Smithsonian Institution, 1984, p. 27). Here is a user now articulated as an ‘individual, at any terminal in the Institution [who] should be able to access any information in the Institution which he or she has the right to see, and update any data which he or she has the right to alter’ (Smithsonian Institution, 1982, p. 2). In contrast to the ‘operator’, here the discourse is characterized by an empathy and sensibility to how the system might be experienced by the user – be they museum employee, specialist or member of the public; ‘The Trust is aware that each individual is unique’, explains one institution’s Policy on Access, appended to its description of a new piece of digital design, ‘both in terms of how he or she learns and in their access needs’ (York Archaeological Trust, 2009). This is a language of ‘user’ that acknowledges not only that there are other systems outside that of the museum that may ‘be in use amongst users’ (Windsor and Maidenhead, 2005), but that these diverse users have themselves different behaviors with these said systems. Here, for instance, the museum begins to recognize the importance of so-called ‘user agents’ – that online users bring their own software that customizes content for that individual’s context, be that their choice of browser, their device, their preferred interface settings, or the assistive technology that will be mediating content to the user (Thackray, 2008; Tyne and Wear 2005). And it is here that we see the museum working from the assumption that, in these new digital settings, ‘users are able to choose’ (IWM, 2009; Signly, 2017b).

With individual preferences and diverse behaviors, as well as an acknowledged existence outside of the museum’s system, the ‘digital user’ is now not just articulated within a context of operational effectiveness (as it was for the user as ‘operator’), but now, also, within a new vivid context of ‘user experiences’ (Wakefield Library, 2006). Consequently, rather than to the operation of the *system*, we hear this discourse of the ‘individual’ user looking instead to the *interface*. These are interfaces that differentiate between (and re-assemble for) the ‘screenreader user’, ‘keyboard only user’ and ‘voice recognition software user’ (British Museum, 2008), and that offer ‘a number of display options [...] which enable users’ (Tate, 2006). It is design that foregrounds an ‘interface to accommodate the needs of people’ (The National Trust, 2008). In this more individualized formation of the ‘digital user’, the emphasis shifts to how the system will work for the user rather than the organization, with the dominant discourse now on ‘usability’ rather than process. In this usage, we see the language of the museum underscoring ‘consultation with users’, and how a ‘great deal of work has been undertaken on usability’ (Tyne and Wear, 2005). With the ‘user’ central to the offer, digital provision is presented as something that ‘allows users to search’, ‘allows users to browse’, ‘allows users to zoom in on any part of an image’; the aim is ‘to provide opportunities to users’, ‘users can find out more’, ‘users can explore a theme’, and ‘users can have fun’ (Tyne and Wear, 2005).

Significantly, in this construction, we also see this ‘digital user’ expressed as part of an on-going evaluative relationship and dialogue; ‘a reciprocal relationship’, as the Canadian Museum for Human Rights puts it, ‘whereby the Museum and the visitor mutually inform each other’ (Jodi Awards, 2015). ‘Going forward’, another explains, ‘we will continue to encourage feedback and consider the views of our users to re-evaluate any aspects that could improve accessibility and usability on this and future online resources’ (National Archives, 2007). This is the museum involving ‘users at each stage’ (Wolverhampton Arts and Museum Service, 2007). This is ‘the needs of the user now driving the design’ Thackray Museum (2008). Indeed, in this discourse, ‘usability’ (and the user at centre of the design) is so significant that it can become the authenticating measure of digital provision; ‘[t]he app has already been updated and continues to be improved and expanded regularly’, one award-winning application explains, ‘with continued input from user groups to ensure the integrity of the project is upheld’ (Jodi Awards, 2017a). Here we hear the ‘integrity’ of the museum’s digital provision reliant upon the involvement (and endorsement) of its digital user – no longer just an operator, but now an individual, with individual needs, and an individual point-of-view.

The User as ‘Actant’

The MDA archive contains a copy of an ‘Open Letter’ written in 2008 by museum access consultant Marcus Weisen setting out a (Roosevelt-inspired) ‘New Deal’ for cultural equality for the UK. The letter formed the basis of a keynote address – delivered two weeks later – to a gathering at the British Museum of prominent accessibility and disability practitioners, policy-makers and academics – including the National Trust, English Heritage, British Library, National Archives, University of Cambridge, National Museums Scotland, and the Wellcome Trust. In the wider history of cultural access and rights, and the role of UK public institutions within this domain, Weisen’s rallying cry for ‘a complete sea-change to let the creative wind of opportunity freely blow’, stands as a key moment

of challenge and collective self-reflection (Weisen, 2008, section 4). However, for tracing our own ideation history here, Weisen's letter also encapsulates the re-making of the 'user' that was taking place at that time within the museum sector. One significance of the letter and its impassioned call by an individual with experience both with policy (in central government) and with practice (with the Royal National Institution of Blind People), was the extent to which the 'user' was relocated into a context of social purposefulness and redefined as an individual with 'cultural freedoms' and agency in the world. The subject of disability had provided the means to develop a much wider re-imagining of the 'user', and the 'digital user'. What Weisen was calling out (shared at a national event that was reflecting on digital media in the context of disability) was what we might style as a shift in the usage of the word 'user' from 'individual' to 'actant'.

Just five years earlier, in her landmark report to the UK museum sector, we can see Annie Delin (2003, pp. 6, 14) identifying a comparable transition: in her case, from the doctrine of equal opportunities (positioning disabled people as a group 'not to discriminate' against), to the 'notion of disability rights (defining disabled people – varied and non-homogenous – as 'people with something to contribute'). What Delin described in the context of the 'user', and that Weisen had later located in the context of the 'digital user', was a 'view of disabled people as valuable citizens with equal rights, who make their own life decision and choices', of 'disabled people as part of society and influential on its progress' (Delin, 2003, p. 7). This was a view of 'users' not just relevant to disabled people, but with universal application. It was, in other words, a language of 'use' (of agency, of empowerment, of affect) that came to characterize a new discourse on the museum's 'user' and 'digital user'. It was a language of agency (of acknowledging the 'digital user' as an empowered person in the world) that we start to see influencing a values-led approach to a universal design ethic within the museum sector from around the turn of the century (Petrie, King & Hamilton, 2005, p. 10). These are the actants – such as the bloggers identified in the Editorial Policy of arts organization 'Disability Arts Online' – who profile their own work '[w]ithin a global context' and produce their own 'quality writing' on Deaf and Disabled-led arts and culture on their own pages within the organization (DAO, 2009; 2009b). These are the actants, in the 'Speaking for Ourselves' project, whose volunteered 'life-stories' heard in 240 hours of recorded testimony on 'living with cerebral palsy' (Scope, 2009). And these are the actants, people with learning disabilities and with agency, who in the case of the Outside In Pathways (2008) worked with the V&A (London), producing their own 'digital artefacts', using the museum 'as a base for the creative activity'. In this more recent construction of the 'digital user', the driver crucially is no longer *efficiency* (of the 'operator'), or *accessibility* (of the 'individual'). Instead it is *agency* that defines the 'user' – a socially active user, within a wider world.

Seeing the Museum through its 'Digital User'

Our three users (the 'operator', the 'individual' and the 'actant'), evidenced through these key archives of professional practice over the last fifty years, help us to understand a shifting frame of reference around museums and digital (see Figure 13.1). In the 'operator' (from the 1960s-70s onwards) we see a generic user seen anonymously as part of the efficiency of the 'designed physical system' (Bai & Lindberg, 1999, p. 794) – set within new standards around operational effectiveness. In the 'individual' (from the 1980s-90s onwards) we see a unique user acknowledged as being outside of the designed physical system with their own personal accessibility needs – amidst new priorities around user experience. And in the 'actant' (from the 2000s-10s) we see the user accepted as socially active and with agency in the wider world – cast within new emphases around social value.

Across these different constructions of the 'digital user', we can also track a change in emphasis in the profession's dominant discourse: from automation and process; to personalization and usability; to empowerment and affect. And as we do, what we also note is the museum at each point reframing its proposition for this user: for the 'operator', raising questions of *how* the system will work; for the 'individual', questions of by *whom* the interface will be used; and for the 'actant' questions of *why* and *where* on the network there will be effect. It is an evolving construction, from functions, to interactions to actions. And a moving center – from *usefulness*, to *use*, to *purposefulness*.

The 'digital user' as ...

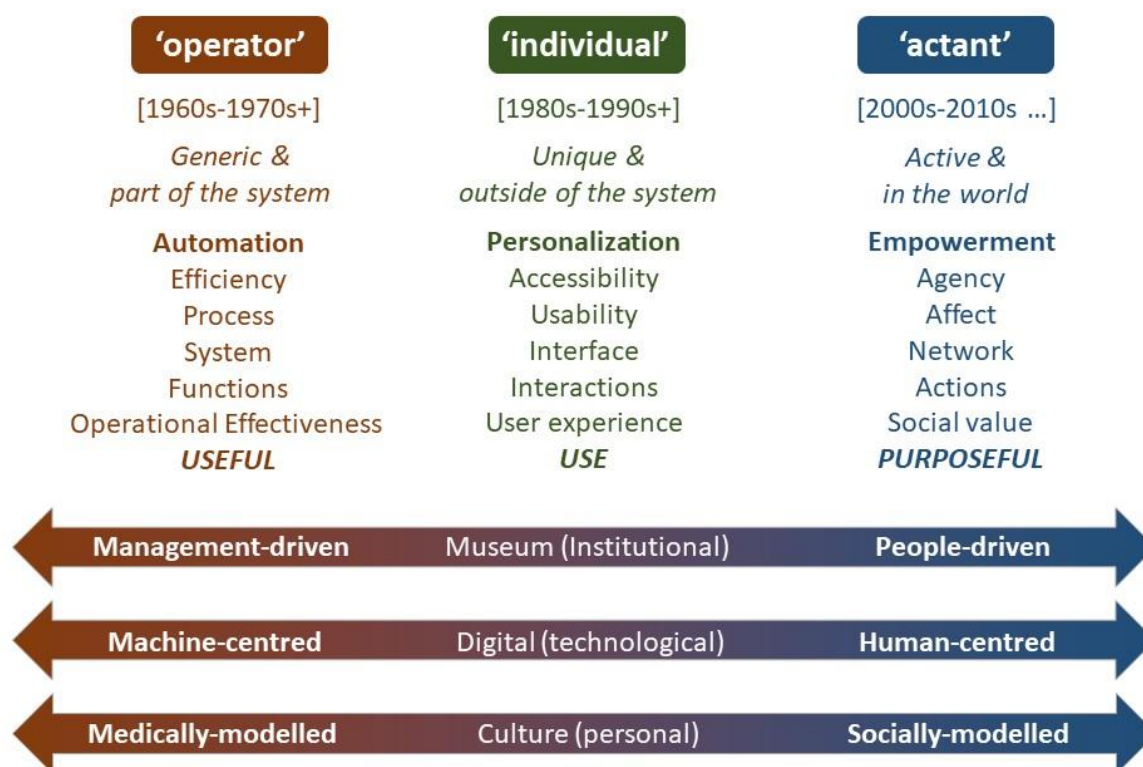


Fig 13.1 An evolution of the museum's 'Digital User'

However, there are wider implications of this evolutionary map. This genealogy of terms also helps us to trace deeper influences that are at play in the museum's construction of its 'digital user'. For behind and within these roles of 'operator', 'individual' and 'actant', we can also hear more profound shifts in the assumptions being made about *machine*, *body* and *organization* – and the relationship between each. After all, the discourse of the 'operator' is one built around machine-centred language of technology, a medically-modelled philosophy of the body, and a management-driven approach to the organization. Whereas, in contrast, the discourse of the 'actant' is one emerging from human-centered language of technology, a socially-modelled (Ellis & Kent, 2011, p. 3) and affirmative (Cameron, 2014, p. 29) philosophy of the body, and a people-driven approach to the organization. In other words, reflected in the evolving construction of this 'digital user' we can find an analogue not just of an evolving museum (and museology), but an evolving society (and sociology). The shift from 'operator', to 'individual', to 'actant' can be understood as a story of growing institutional maturity and confidence around digital, but – more meaningfully perhaps – it is also the story of: design practice becoming more user-centered; of an inclusive society respecting human rights; and of the institution of the museum becoming more socially purposeful. Plainly put, in the making and re-making of the museum's 'digital user', we see a familiar transition – from the system at the center, to the person at the center.

If this model of 'user' and 'use' is to serve us as a critical framework (here and forwards), there are perhaps other important characteristics of it to be noted in our conclusions, specifically concerning: the chronology and sequencing of these 'user' terms; the importance of context in using this as a term in motion; the specificity of these 'user' constructions to the museum in particular; and the catalytic role for critical disability studies (and museums' experience of disability) within our on-going understanding of the 'digital user'. First, it is important to recognize that whilst our three (re-)makings of the 'digital user' evidenced here are largely *sequential*, they are also *enduring*. In other words, though the prevalence of each can be located in particular moments in museums' recent history (be that the 1960-70s *era of automation*, the 1980s-90s *era of personalization*, or modern *era of empowerment*), and though each may resonate within particular generations of practice, each of them did not supplant and supersede the former. In short, rather than replacing each other, these constructions have multiplied and co-existed alongside each other. Today our digital user is still constructed at times as 'operator' and as

‘individual’, as well as styled more contemporaneously as ‘actant’. And, as a concept, we suspect the ‘digital user’ will remain – in this way – in motion.

Second, as we look back and continue to write (and learn from) histories of digital cultural heritage, we can no longer assume that the term ‘user’ is a constant. We cannot project back on to earlier uses of the word the modern inferences and referents associated with it. Similarly, as well as a changing meaning through time, there are clearly different contexts of use – technologically, managerially and curatorially. By building an ideation history, as we have here, we are now sensitized to the importance of sustaining a precision in our professional and scholarly lexicon and retaining this awareness to these complex and overlaying (media, business and design) contexts for ‘user’ and ‘use’.

Third, although there are many features of this ideation history of ‘the user’ that other disciplines and professions may recognize and share (the origins within operational efficiency; the shift to a user-centered perspective; the rise of the user ‘in the world’), the genealogy mapped here, and the user ‘types’ patterned here, ought to be seen to belong specifically to a *museum* context. Owing to museums’ unique history, and as a consequence of the specific nature of the sector, and due to the particular formations of the museology over the last fifty years, there are nuances and subtleties here (curatorial, managerial, societal) that belong to museums alone. This particular (re-)making of the digital user speaks, therefore, specifically to the museum. Consequently, this reminds us of the situated nature of media and digital studies, and that whilst we may learn from other related subjects and disciplines, so we also ground their ideas within our own professional and scholarly context. Whilst this account of digital practice would not in itself purport to being ‘ethnographic’ in a disciplined sense (see Geismar, 2012), it does nonetheless stand to demonstrate the (postdigital) importance of reflecting on the ‘digital user’ critically, historically and in a culturally situated way.

Fourth, and finally, we note here that just as ‘museum’ has helped us to think about ‘digital user’ (as a unique stage on which ideas of medialization can be explored), and ‘digital’ has helped us to think about the ‘museum’ (and its changing constructions of its audiences and users), so we also notice how ‘disability’ has helped us here to think about ‘digital’. After all, as the archive of the Jodi Mattes Trust has shown so vividly, the place to follow the story of the ‘digital user’ in the museum, is in the museum’s provision for the disabled user. Through this archive of aspiring accessible digital design, we can see the experience of the disabled visitor animating, magnifying and amplifying the evolution of the ‘user’ from ‘operator’, to ‘individual’, to ‘actant’. The history of accessible digital design in the museum is where our history of the ‘digital user’ is most clearly seen. Furthermore, in the museum’s maturing views and values around disability (as manifest in the growing confidence, insight and creativity of the archive’s ‘nominations’), we witness not just a *consequence* but also a *cause* of the evolving construction of the ‘digital user’. In other words, just as the emergence of the user as ‘individual’ and then as ‘actant’ drives a more affirmational and accessible approach to digital design (‘sensitised to the manifold complexities of the body’ (Imrie, 2014, p. 288)), so then also the museum’s growing experience of disability drives (*and expects*) the construction of the user as a socially active, empowered agent in the world. In short, as we look through the Jodi Mattes Trust archive we are not only seeing the *results* of the museum constructing the user as ‘individual’ and then as ‘actant’ – but also the *causes* of it. It was its experience of disability that helped the museum see the user as ‘actant’.

In this way, our ideation history of the ‘digital user’ leaves us perhaps with an even more profound outcome – one in which disability provides a powerful frame through which the ‘user’ and ‘digital user’ might be re-approached. This is a perspective from which disability is not merely something to be understood, provided for, and studied (see for instance, French & Swain, 2014), but is instead an experiential and intellectual way of framing analysis on ‘user’ and ‘use’. This can be a frame informed by the counterdiscourse reasoning of critical disability studies. Faye Ginsburg, for instance, shows us how, within the study of disability, digital technology can be seen to provide ‘unanticipated and powerful platforms’ for alternative self-representation, but it can do so ‘in ways that expand our collective sense of personhood and publics’ (Ginsburg, 2012, pp. 102–103). This is a critical perspective from which the disabled experience offers ways of constructing the world, challenging particular assumptions about citizenship, and not accepting assigned responsibilities. This is not just about seeking to ‘rehabilitate the relevance of impairment to a social, cultural, and political investigation of disability, particularly within the context of digital technologies’ (Ellis & Kent, 2011, p. 4), but it is about disability (not just for those of us with impairment) as a critical frame, not as an object of gaze, but as affirmative lived experience from which to be activist (Sheldon, 2014, p. 331). But, moreover, crucially, compellingly, this is the museum’s essential conceptualization of the ‘digital user’, animated, agitated and activated by insights and experiences of disability.

Conclusion

As a subject and as practice, 'museum computing' tends (still) to center technology. It is technology, after all, that has driven institutional and sectoral change, that has redefined roles and relationships between organization and audience, and that has enabled the re-assembling (sometimes dramatically so) of multiple concepts, processes and practices of 'museum'. The attraction – perhaps understandably with so much in transition, so much at stake, and so much to say – has been to call out the next big inventive step in systems and device, and to spotlight the next audacious leap in practice and platform. However, amidst this orientation around technology, the *practice of digital* risks eclipsing the *meaning of digital*.

And yet today – especially with fifty years of digital practice now behind us on which to reflect – there is an opportunity for other ways of looking at digital in the context of the museum, and other ways of formulating equations in which digital might function. There are alternative framings to our subject that do not necessarily have to accept the orthodox focus on the functionality of the technology, or changes to business processes, or even the characteristics of digitally-enabled experience. We can, in short, choose to decline the technocratic and technocentric, and instead allow digital to become a means for other shapes of conversation, other lines of enquiry, and (even) allow other roles for digital in our discourse. This is digital not just as an object to be understood in itself, but as a critical lens and proxy through which to understand other (non-digital) things. This is digital not just within the frisson of the present and emerging future but set in a longer historical view. This is digital not just as a technology, but as an idea.

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